REAL-TIME MONITORING TREN DEDOLARISASI DAN DAMPAKNYA TERHADAP STABILITAS MATA UANG ASEAN

Proposal Big Data Project

Disusun Oleh Kelompok 4 :

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TEKNIK INFORMATIKA

FAKULTAS TEKNOLOGI INFORMASI DAN SAINS DATA

UNIVERSITAS SEBELAS MARET

SURAKARTA

2026

ARSITEKTUR SISTEM

Alur Data End-to-End

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┌────────────────────────────────────────────────────────────────────────────────┐
│                           ALUR DATA END-TO-END                                  │
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│                                                                                │
│  LAPTOP 1 (Jimly)          LAPTOP 2 (Nadhifa)          LAPTOP 3 (Adrian)       │
│  ┌───────────────────┐     ┌───────────────────┐       ┌────────────────────┐  │
│  │  Yahoo Finance    │     │  Apache Spark     │       │  Apache Cassandra │  │
│  │       ↓           │     │       ↓           │       │  Elasticsearch    │  │
│  │  Python Producer  │─────→│  Preprocessing    │       │  FastAPI          │  │
│  │       ↓           │ Kafka│       ↓           │       │  React Dashboard  │  │
│  │  Apache Kafka     │     │  Feature Eng.     │       │                   │  │
│  │  Zookeeper        │     │       ↓           │       │                   │  │
│  └───────────────────┘     │ K-Means, DBSCAN,  │       │                   │  │
│   Data Ingestion Layer     │ AHC               │──────→│  Serving Layer    │  │
│                            └───────────────────┘       └────────────────────┘ │
│                               Processing Layer                                 │
└────────────────────────────────────────────────────────────────────────────────┘
```

Komunikasi Antar Laptop

Dari	Ke	Media	Data	Frekuensi
Laptop 1 (Jimly)	Laptop 2 (Nadhifa)	Kafka topic `forex-raw`	OHLC + timestamp	Setiap 60 detik
Laptop 2 (Nadhifa)	Laptop 3 (Adrian)	Kafka + Cassandra write	Features + cluster labels	Setiap micro-batch
Laptop 3 (Adrian)	Browser / User	HTTP REST + WebSocket	JSON hasil query	On-demand + push

PEMBAGIAN TUGAS

No	Anggota	Laptop	Layanan	Tanggung Jawab & Output
1	**Jimly Syahbatin** (L0224033)	**Laptop 1** *Data Ingestion*	• Apache Kafka & Zookeeper • Python Kafka Producer • Yahoo Finance Fetcher	• Menulis kode producer Python untuk fetch data forex dari Yahoo Finance secara real-time • Konfigurasi Kafka topics, partisi, dan retention • Menyediakan data streaming ke Laptop 2 • **Output:** Data mentah kurs di topic `forex-raw`
2	**Nadhifa Sakha Tri Yasmin** (L0224036)	**Laptop 2** *Processing*	• Apache Spark Structured Streaming • Model ML: K-Means, DBSCAN, AHC	• Membaca data Kafka dengan Spark Streaming • Preprocessing & feature engineering (rolling window, korelasi DXY/CNY, RSI, volatilitas) • Implementasi 3 algoritma clustering • **Output:** Fitur di topic `forex-features` + hasil clustering di `clustering-results`
3	**Adrian Farrel Aziz Yatyoga** (L0224040)	**Laptop 3** *Serving Layer* **(Terberat)**	• Apache Cassandra • Elasticsearch • FastAPI • React Dashboard	• Setup & konfigurasi Cassandra + Elasticsearch • Membangun REST API dengan FastAPI (9 endpoint) • Membangun dashboard React (scatter plot, dendrogram, tabel outlier) • **Output:** API endpoint + dashboard visualisasi real-time

Catatan: Laptop 3 (Adrian) mendapat beban **paling berat** karena menjalankan 4 service besar secara simultan: Cassandra (storage time-series), Elasticsearch (heap JVM min 1GB), FastAPI (server API), dan React dashboard. Dibutuhkan laptop dengan RAM ≥ 16 GB dan storage SSD ≥ 50 GB.

ROADMAP IMPLEMENTASI

Proyek dibagi menjadi 6 fase pengerjaan. Setiap fase bisa diuji secara independen sebelum lanjut ke fase berikutnya.

Fase 0: Persiapan Lingkungan (1 hari)

PIC: Semua anggota

1. Buat struktur folder proyek bersama di GitHub
2. Install dependencies: `pip install kafka-python yfinance pandas pyspark fastapi uvicorn cassandra-driver elasticsearch`
3. Setup Docker Engine di ketiga laptop
4. Buat script inisialisasi: `create-kafka-topics.sh`, `init-cassandra.cql`, `init-elasticsearch.sh`

Fase 1: Data Ingestion — Laptop 1 (2 hari)

PIC: Jimly Syahbatin

File: `producer/config.py`, `fetcher.py`, `kafka_producer.py`, `main.py`

- Fetch 9 ticker paralel: IDR, THB, SGD, MYR, PHP, VND, DXY, CNY, Gold
- Kirim ke Kafka topic `forex-raw` format JSON tiap 60 detik
- Mock data generator untuk development awal

Verifikasi: Konsumen Kafka bisa membaca data dari topic `forex-raw`.

Fase 2: Storage — Laptop 3 (1 hari)

PIC: Adrian Farrel Aziz Yatyoga

Cassandra DDL:

```
CREATE KEYSPACE dedolarisasi WITH replication = {'class': 'SimpleStrategy',
'replication_factor': 1};

CREATE TABLE forex_rates (
    currency_pair text, ts timestamp,
    open double, high double, low double, close double, volume bigint,
    PRIMARY KEY ((currency_pair), ts)
```

```
) WITH CLUSTERING ORDER BY (ts DESC);

CREATE TABLE features (
    currency_pair text, ts timestamp,
    returns_1d double, rolling_mean_5d double, rolling_mean_20d double,
    rolling_std_5d double, volatility_20d double,
    corr_dxy_20d double, corr_cny_20d double,
    PRIMARY KEY ((currency_pair), ts)
) WITH CLUSTERING ORDER BY (ts DESC);

CREATE TABLE clustering_results (
    batch_id text, ts timestamp, algorithm text,
    currency_pair text, cluster_label int, cluster_name text,
    is_outlier boolean, silhouette_score double,
    PRIMARY KEY ((batch_id), ts, algorithm, currency_pair)
) WITH CLUSTERING ORDER BY (ts DESC);
```

Verifikasi: Data bisa ditulis dan dibaca dari Cassandra & Elasticsearch.

Fase 3: Spark Streaming — Laptop 2 (3-4 hari)

PIC: Nadhifa Sakha Tri Yasmin

3a. Stream Reader (1 hari)

Baca Kafka `forex-raw`, parse JSON, simpan ke Cassandra `forex_rates`.

3b. Feature Engineering (1-2 hari)

Fitur	Rumus	Window	Tujuan
returns_1d	(close - prev) / prev	1	Return harian
log_return	ln(close / prev)	1	Return kontinu
rolling_mean_5d/20d	Moving average	5 / 20	Tren jangka pendek/menengah
rolling_std_5d/20d	Std dev returns	5 / 20	Volatilitas lokal
volatility_20d	Std dev log_return	20	Volatilitas klasik
corr_dxy_20d	Korelasi vs DXY	20	**Ketergantungan USD**
corr_cny_20d	Korelasi vs CNY	20	**Kedekatan Yuan**
rsi_14	Wilder's RSI	14	Overbought/oversold
bb_upper/lower	MA $\pm$ 2σ	20	Bollinger Bands

3c. Clustering (1 hari)

Input vektor (3D): `[corr_dxy_20d, corr_cny_20d, volatility_20d]`

- **K-Means** — PySpark ML. Label: Pro-Dollar / Transisi / Mendekati Yuan
- **DBSCAN** — sklearn via Pandas UDF. Label -1 = outlier anomali
- **AHC** — scipy, batch periodik 6 jam. Output dendrogram

Verifikasi: Hasil clustering muncul di Cassandra `clustering_results`.

Fase 4: FastAPI — Laptop 3 (2 hari)

PIC: Adrian Farrel Aziz Yatyoga

Method	Path	Sumber

GET	`/api/v1/rates/{currency}`	Cassandra
GET	`/api/v1/rates/{currency}/history?range=7d|30d|90d`	Cassandra
GET	`/api/v1/features/{currency}`	Cassandra
GET	`/api/v1/clusters/latest?algorithm=kmeans|dbscan|ahc`	ES / Cassandra
GET	`/api/v1/clusters/summary`	Elasticsearch
GET	`/api/v1/dashboard/summary`	Cassandra + ES
WS	`/api/v1/ws/clusters`	Kafka → push

Fase 5: React Dashboard — Laptop 3 (3 hari)

PIC: Adrian Farrel Aziz Yatyoga

Route	Visualisasi
`/`	Summary cards, real-time ticker, pie chart cluster, line chart DXY vs ASEAN
`/clustering`	Scatter plot interaktif (corr_dxy vs corr_cny), warna per cluster
`/dendrogram`	D3.js dendrogram interaktif dari AHC
`/anomalies`	Tabel outlier DBSCAN + time series anomaly highlight

Stack: React 18 + TypeScript + Vite + Recharts + D3.js + Tailwind CSS + WebSocket.

Fase 6: Integrasi & E2E (2 hari)

PIC: Semua anggota

1. Nyalakan Docker Compose di 3 laptop
2. Verifikasi alur: Laptop 1 → Kafka → Laptop 2 → Cassandra/ES → Laptop 3
3. Tunggu rolling window 20 data point terpenuhi
4. Verifikasi cluster muncul di dashboard
5. Uji WebSocket real-time update

LEMBAR PENGESAHAN

Proposal Big Data Project ini telah disusun oleh Kelompok 4 dan disetujui untuk dilaksanakan.

Surakarta, Juni 2026

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